LAB 3 [Group # → 9 to 16]

[Platform → HACKERRANK]

Introduction To Programming

(24-01-22)

SOLUTIONS

Q1 Bamboos

```
n = int(input())
lst = []
for i in range(n):
    x = int(input())
    lst.append(x)
lst = sorted(lst)
# print(lst)
mid = n // 2
sm = 0

for i in range(n):
        sm += abs(lst[i] - lst[mid])
print(sm)
```

Q2 Sam is back

```
n = int(input())

lst = []
for i in range(n):
    lst.append(int(input()))
tot = 0
for x in lst:
    tot +=x
left = 0
isPos = False;
for i in range(0,n-1):
    left += lst[i]
    if left == tot - left :
        isPos = True
        break
if isPos :
    print("YES")
else :
    print("NO")
```

Q3 Minutes passed from the midnight

```
hr = int(input())
mns = int(input())

def is_valid(hr, mns):
    if hr>24 or hr<0:
        return "Invalid"
    if mns>60 or mns<0:
        return "Invalid"

    return "Valid"

validity = is_valid(hr, mns)

if validity == "Valid":
    print(hr*60 + mns)
else:
    print(validity)
```

Q4 Is it perfect?

```
num = int(input())

def is_perfect(num):
    s = 0
    for i in range(1,num):
        if num%i ==0:
            s +=i

    if s==num:
        print("Perfect")
    else:
        print("Not Perfect")
    return

is_perfect(num)
```

Q5 It's the salary day

```python
basic = float(input())

gross, hra, da = 0, 0, 0

if basic<=10000:
    hra = 0.09*basic
    da = 0.11*basic
elif basic>10000 and basic <=20000:
    hra = 0.18*basic
    da = 0.22*basic
else:
    hra = 0.27*basic
    da = 0.33*basic
gross = basic +da +hra

print(f'{gross:.2f}')
```

Q6 Population Growth

```python
def time_taken(a0,p,n,t):
    a = a0
    time = 0
    while t > a:
        a += (p/100)*a + n
        time += 1
    return(time)

a0 = int(input())
p = float(input())
n = int(input())
t = int(input())
print(time_taken(a0,p,n,t))
```

Q7 Sum of numbers

```python
a = int(input())
b = int(input())

def findSum(a,b):
    #good luck!
    if a==b:
        return a
    s = 0
    if a<b:
        for x in range(a, b+1):
            s += x
        return s

    if a>b:
        for x in range(b, a+1):
            s += x
        return s
print(findSum(a, b))
```